



Manifold for Type 1 Diabetes – Restoration

Diabetes Research Institute, University of Miami — long-horizon islet-transplant registry analytics and BioHub translation

1 · THE CLINICAL PROBLEM

The Diabetes Research Institute holds an asset that no other institution can replicate: the longest continuous programmatic record of clinical islet transplantation in the world, underpinned by the Ricordi Chamber isolation methodology that made the field's clinical arm reproducible. The DRI record spans:

- **Multi-decade allogeneic islet-transplant registry.** Donor quality, isolation yield and viability, recipient HLA and autoantibody profile, immunosuppression regimen exposure, C-peptide AUC trajectory, insulin-dependence status, and long-horizon graft-survival outcomes across a patient population no other center matches.
- **Edmonton Protocol and post-Edmonton regimen variants.** Longitudinal evidence base on immunosuppression-regimen modifications and their causal effect on graft durability — a natural domain for heterogeneous-treatment-effect inference and Pearl-do-calculus framing.
- **BioHub and encapsulation programs.** Next-generation islet-delivery architecture — the omental biological scaffold, encapsulation strategies, and their interaction with continuing autoimmune and alloimmune pressure — moving from preclinical to clinical with dense preclinical signal.

Three capabilities remain underdeveloped in current analytic practice across islet-transplantation registries, including analyses applied to the DRI record to date:

- Unified causal modeling of donor quality, recipient state, immunosuppression exposure, and long-horizon graft outcome — recovering the directed structure rather than pursuing marginal regression across each covariate independently.
- Long-horizon graft-survival hazard decomposition with calibrated uncertainty — parsing the subject-level features driving decade-plus durability from the features driving early attrition.
- Principled immunosuppression-regimen and BioHub-variant prioritization on a unified causal graph — rather than regimen-by-regimen empirical iteration.

The DRI record is a unique substrate: the longest continuous clinical-islet-transplant archive in the world, paired with an active BioHub translational program. The question is no longer whether islet replacement can produce insulin-independence — the DRI record demonstrates it — but which donor, recipient, regimen, and delivery-architecture combinations sustain the outcome over a decade and beyond. That is a problem of causal inference on a uniquely long-horizon dataset.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities form the basis of the DRI-configured deployment:

1. **Causal discovery (Spirtes / PC algorithm) on the DRI transplant registry.** Recovers the directed structure among donor quality, isolation yield, recipient HLA and autoantibody profile, immunosuppression exposure, and long-horizon graft outcome — the substrate for principled heterogeneous-recipient and heterogeneous-regimen inference.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Individual estimators are configured to registry-scale longitudinal data — long-horizon survival models, nonlinear mixed-effects C-peptide decay, heterogeneous-treatment-effect learners on regimen variants, Bayesian change-point on graft-compromise timing — with the composite score providing a stable integration layer.
3. **Pearl do-calculus and interdiction.** Regimen and delivery-architecture prioritization — including BioHub-variant design — is framed as a maximum-attenuation intervention on the inferred causal graph, producing principled ranking of modifications rather than empirical sequential iteration.

3 · OF PILLAR MAPPING - DRI ISLET-TRANSPLANT PHYSIOLOGY

OF PILLAR	DRI REGISTRY READING
I — Interaction	Donor quality × recipient HLA × immunosuppression × autoantibody profile interactions; BioHub substrate × host-tissue interaction
R — Reserve	Post-transplant functional graft mass (C-peptide AUC on MMTT); residual graft mass at long horizons; isolation-yield-normalized engraftment reserve
J — Junction	Immune-metabolic coupling across allo- and auto-immunity; cytokine-panel ⇔ C-peptide transfer entropy; BioHub scaffold × host-immune coupling
C — Cascade	Graft-attrition cascades: allorejection, autoimmunity recurrence, chronic-immunosuppression toxicity; BioHub-specific fibrotic-response cascades
T — Temporal	Hazard of graft attrition at 1, 5, 10, and 15-year horizons; time-to-insulin-dependence return given trajectory

4 · CRITICALITY ESTIMATOR CONFIGURATION

DRI registry data is registry-scale and long-horizon, with heterogeneous regimen exposure and dense donor-recipient covariate structure. The criticality layer is configured accordingly:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Long-horizon survival models (Cox, DeepSurv, time-varying covariate survival)	Graft-attrition hazard at 1, 5, 10, and 15-year horizons given donor, recipient, and regimen covariates	R, T
Heterogeneous-treatment-effect learners (Causal Forest, BART, S / T / X-learners) on regimen variants	Immunosuppression-regimen response differentiation across recipient subpopulations	I, C
Nonlinear mixed-effects C-peptide decay models	Per-subject post-transplant function trajectory; early-attrition signature detection	R, T
Bayesian online change-point (BOCPD) on post-transplant metabolic and inflammatory signals	Graft-compromise onset detection with calibrated uncertainty; BioHub-specific fibrotic-response onset	J, T

Additional estimators available within the same layer for subsequent phases:

- **Transfer entropy** between autoantibody titers, inflammatory cytokine panels, and C-peptide trajectories — quantifies immune-metabolic coupling (J) driving long-horizon graft attrition.
- **Persistent homology** on donor-recipient covariate manifolds — topological regime detection across graft-outcome strata, valuable for identifying durable-response phenotypes.

5 · PROPOSED INITIAL ENGAGEMENT (PHASE 1)

The scope of the proposed Phase 1 engagement is defined to produce a deliverable of demonstrable analytic value to the DRI within a contained time and data footprint.

- **Primary dataset.** DRI allogeneic islet-transplant registry: donor and recipient covariates, isolation yield and viability, immunosuppression regimen, longitudinal C-peptide AUC, insulin-dependence status, and graft-outcome at multi-year follow-up. Where available, retrospective BioHub preclinical and early clinical signal as complementary input.
- **Access path.** DRI-internal Data Use Agreement and IRB, under the sponsorship of a DRI-affiliated principal investigator.
- **Deliverable.** Per-subject Ω F trajectory integrating long-horizon survival, heterogeneous-treatment-effect, and change-point estimators. Graft-attrition hazard decomposition at 1, 5, and 10-year horizons. Pearl-do-calculus ranking of immunosuppression and delivery-architecture modifications predicted to most attenuate long-horizon attrition.
- **Success criteria.** Long-horizon hazard model with calibration and discrimination matching or exceeding the DRI's pre-specified biostatistical analyses; identification of at least one durable-response subgroup decomposable along Ω F pillars; interdiction-engine regimen and BioHub-variant rankings interpretable and actionable by the DRI clinical team.
- **Timeline.** Twelve to fourteen weeks from data access.
- **Excluded from Phase 1** (reserved for the Phase 2 prospective arm): prospective BioHub or encapsulated-variant trial design, live CGM ingestion, bespoke immunologic assays.

6 · RATIONALE AND TIMING

- **Long-horizon data has no substitute.** The DRI registry is the only islet-transplant archive of its duration and continuity. The analytic tools appropriate to long-horizon multi-covariate data — causal discovery, deep survival, heterogeneous-treatment-effect inference — have only recently matured to production grade against it.
- **BioHub is at the translational inflection point.** The omental-scaffold, encapsulated, and next-generation delivery programs moving through the DRI translational pipeline have preclinical and early clinical signal; their prioritization is directly a causal-inference problem.
- **Replacement paradigm broadly has crossed proof-of-principle.** Insulin-independence in early-phase Vertex VX-880 cohorts reinforces the field's focus on generalization — which donor, recipient, regimen, and architecture combinations sustain the outcome — which is the question the DRI registry is uniquely positioned to answer.
- **Relevant funding mechanisms are active.** The Breakthrough T1D Cures program, the Helmsley Charitable Trust Restoring Insulin Production stream, and NIH NIDDK all have active Requests for Proposals spanning islet-transplantation analytics.
- **The platform is production-ready.** Manifold's causal, criticality, and interdiction engines are production-grade; configuration for the DRI registry and BioHub domains represents bounded, well-scoped work.

7 · PROPOSED PARTNERSHIP

Partnership with a Diabetes Research Institute principal investigator at the University of Miami Miller School of Medicine is sought for:

1. Retrospective DRI islet-transplant registry and BioHub data access under the DRI's existing Data Use Agreement and IRB.
2. Clinical co-authorship on the long-horizon graft-survival and regimen-interdiction benchmark manuscript.
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK for the Phase 2 prospective arm.

In exchange, Apex Analytica contributes the causal-inference engine, the Ω F framework, and the engineering team. Apex retains commercial rights to the platform; research intellectual property from the joint work is shared on standard academic-industry terms.